Answer Key:

1. **D**
2. **A**
3. **B**
4. **C**
5. **D**
6. **D**
7. **E**
8. **B**
9. **D**
10. **C**
11. **A**
12. **A**
13. **D**
14. **C**
15. **C**
16. **A**
17. **D**
18. **C**
19. **A**
20. **B**
21. **E**
22. **A**
23. **B**
24. **B**
25. **C**
26. **B**
27. **C**
28. **A**
29. **D**
30. **D**

Solutions:

1. $\boxed{\textbf{D}}$: $\left|\begin{bmatrix}2 & 0\\ 2 & 6\end{bmatrix} - \begin{bmatrix}0 & -2\\ 2 & -6\end{bmatrix}\right| \implies \left|\begin{bmatrix}2 & 2\\ 0 & 12\end{bmatrix}\right| \implies 2(12) - 2(0) = \boxed{24}$.

2. $\boxed{\textbf{A}}$: We have

$$\begin{aligned}
\vec{u} - \vec{w} + 2\vec{v} &= \langle 2,2,-6\rangle - \langle -1,-2,3\rangle + 2\langle 7,7,7\rangle && \text{(Substitution)}\\
&= \langle 2,2,-6\rangle + \langle 1,2,-3\rangle + \langle 14,14,14\rangle && \text{(Scalar Multiplication)}\\
&= \boxed{\langle 17,18,5\rangle}. && \text{(Vector Addition)}
\end{aligned}$$

3. $\boxed{\textbf{B}}$: Observe that the desired result, $\left(\begin{bmatrix}6 & 5 & 2\\ 10 & -10 & 10\\ 7 & 7 & 7\end{bmatrix} + \begin{bmatrix}-1 & 13 & 0\\ -5 & 6 & 3\\ 7 & 7 & 7\end{bmatrix} + \begin{bmatrix}-2 & 0 & -8\\ -5 & 2 & 3\\ 7 & 7 & 7\end{bmatrix}\right)_{23}$, is the element in the 2nd row and 3rd column of the resulting sum. Instead of finding the sum and then looking for the desired result, we can instead just sum the elements in the 2nd row and 3rd column for the 3 matrices. Thus, we have

$$\begin{aligned}
\left(\begin{bmatrix}6 & 5 & 2\\ 10 & -10 & 10\\ 7 & 7 & 7\end{bmatrix} + \begin{bmatrix}-1 & 13 & 0\\ -5 & 6 & 3\\ 7 & 7 & 7\end{bmatrix} + \begin{bmatrix}-2 & 0 & -8\\ -5 & 2 & 3\\ 7 & 7 & 7\end{bmatrix}\right)_{23} &= 10 + 3 + 3\\
&= \boxed{16}.
\end{aligned}$$

4. $\boxed{\textbf{C}}$: We have that $\langle 6,-6\rangle \cdot \langle -1,7\rangle = -48 = \|\langle 6,-6\rangle\|\|\langle -1,7\rangle\|\cos\gamma$. So

$$\begin{aligned}
\left(6\sqrt{2}\right)\left(5\sqrt{2}\right)\cos\gamma &= -48\\
60\cos\gamma &= -48\\
\cos\gamma &= -\frac{4}{5}
\end{aligned}$$

Since γ is the measure of the smaller of the two angles formed by the two vectors, we have that $0 \le \gamma \le \pi \implies \sin\gamma \ge 0$. Thus,

$$\begin{aligned}
\sin^2\gamma + \cos^2\gamma &= 1\\
\sin^2\gamma &= 1 - \left(-\frac{4}{5}\right)^2\\
\sin^2\gamma &= \frac{9}{25}\\
\implies \sin\gamma &= \boxed{\frac{3}{5}}.
\end{aligned}$$

5. $\boxed{\textbf{D}}$: Observe that $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}^2 = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$. We then have that $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}^3 = \begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}$. Similarly, $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}^4 = \begin{bmatrix}1 & 4\\ 0 & 1\end{bmatrix}$.

We can prove, via induction, that $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}^n = \begin{bmatrix}1 & n\\ 0 & 1\end{bmatrix}$. Thus, $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}^{2026} = \boxed{\begin{bmatrix}1 & 2026\\ 0 & 1\end{bmatrix}}$.

6. $\boxed{\textbf{D}}$: For an $n \times n$ nilpotent matrix $\mathcal{N}$, $\mathcal{N}^k = 0_{n\times n}$ for some positive integer $k \leq n$. Since all of the matrices are 2×2, we can conclude that $k = 1$ or $k = 2$.

Since none of the answer choices are $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, we need to check for the case where $k = 2$.

Squaring each of the answer choices, we see that $\begin{bmatrix} 3 & 9 \\ -1 & -3 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. Thus, we have that our answer is $\boxed{\begin{bmatrix} 3 & 9 \\ -1 & -3 \end{bmatrix}}$.

Note: Knowing this matrix is nilpotent will be helpful for Question 25.

7. $\boxed{\textbf{E}}$: Observe that $\begin{vmatrix} 7 & -3 & 1 \\ 1 & 7 & -3 \\ -3 & 1 & 7 \end{vmatrix} = 380 \neq 0$. Since this 3×3 matrix is not singular, its reduced-row echelon form is $\boxed{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$.

8. $\boxed{\textbf{B}}$: The volume of a tetrahedron, for vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$, is $\frac{1}{6}|(\vec{u} \times \vec{v}) \cdot \vec{w}| = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$. Choosing $(7, 7, 7)$ as the tail for our three vectors, we have

$$\begin{aligned} \vec{u} &= \langle 5, 6, 5 \rangle - \langle 7, 7, 7 \rangle = \langle -2, -1, -2 \rangle \\ \vec{v} &= \langle 2, -3, 8 \rangle - \langle 7, 7, 7 \rangle = \langle -5, -10, 1 \rangle \\ \vec{w} &= \langle 8, 9, 10 \rangle - \langle 7, 7, 7 \rangle = \langle 1, 2, 3 \rangle \end{aligned}$$

And,

$$\frac{1}{6}\begin{vmatrix} -2 & -1 & -2 \\ -5 & -10 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \frac{48}{6} = \boxed{8}.$$

9. $\boxed{\textbf{D}}$: By determinant rules, we have $\det \mathcal{G} = \begin{vmatrix} 4 & 5 \\ 1 & 2 \end{vmatrix} \begin{vmatrix} -3 & 7 \\ -5 & 12 \end{vmatrix} \begin{vmatrix} 9 & -8 \\ -6 & 7 \end{vmatrix} \begin{vmatrix} -15 & 9 \\ 12 & -7 \end{vmatrix} \implies 3 \cdot (-1) \cdot (15) \cdot (-3) = \boxed{135}$.

10. $\boxed{\textbf{C}}$: $\begin{vmatrix} -2 & 3 & -2 \\ 1 & 0 & 5 \\ 0 & -1 & -4 \end{vmatrix} = (-2)\begin{vmatrix} 0 & 5 \\ -1 & -4 \end{vmatrix} - (3)\begin{vmatrix} 1 & 5 \\ 0 & -4 \end{vmatrix} + (-2)\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = (-2)(5) - (3)(-4) + (-2)(-1) = \boxed{4}$.

11. $\boxed{\textbf{A}}$: The matrix of minors of $\mathcal{G}$ is $\begin{bmatrix} 5 & -4 & -1 \\ -14 & 8 & 2 \\ 15 & -8 & -3 \end{bmatrix}$. Applying the "checkerboard" pattern,

$$\mathcal{G}_{\mathcal{C}} = \begin{bmatrix} 5 & 4 & -1 \\ 14 & 8 & -2 \\ 15 & 8 & -3 \end{bmatrix}$$

(With $\mathcal{G}_{\mathcal{C}}$ as defined in Question 12)

Therefore,

$$\begin{aligned}
\mathcal{G}^{-1} &= \frac{1}{\det \mathcal{G}}\left(\mathcal{G}_{\mathcal{C}}^{T}\right) \\
&= \frac{1}{4}\begin{bmatrix} 5 & 4 & -1 \\ 14 & 8 & -2 \\ 15 & 8 & -3 \end{bmatrix}^{T} \\
&= \frac{1}{4}\begin{bmatrix} 5 & 14 & 15 \\ 4 & 8 & 8 \\ -1 & -2 & -3 \end{bmatrix} \\
&= \boxed{\begin{bmatrix} \frac{5}{4} & \frac{7}{2} & \frac{15}{4} \\ 1 & 2 & 2 \\ -\frac{1}{4} & -\frac{1}{2} & -\frac{3}{4} \end{bmatrix}}.
\end{aligned}$$

12. $\boxed{\textbf{A}}$: Instead of directly computing $\det \mathcal{G}_{\mathcal{C}}$, we can consider the formula for the inverse of a matrix. We have that $\mathcal{G}^{-1} = \mathcal{G}_{\mathcal{C}}^{T} / \det \mathcal{G}$. Thus,

$$\begin{aligned}
\mathcal{G}^{-1} = \mathcal{G}_{\mathcal{C}}^{T} / \det \mathcal{G} &\implies \det \mathcal{G} \cdot \mathcal{G}^{-1} = \mathcal{G}_{\mathcal{C}}^{T} \\
&\implies \det\left(\det \mathcal{G} \cdot \mathcal{G}^{-1}\right) = \det\left(\mathcal{G}_{\mathcal{C}}^{T}\right) && \text{(Determinant of Both Sides)} \\
&\implies \det\left(\mathcal{G}_{\mathcal{C}}\right) = \det\left(\det \mathcal{G} \cdot \mathcal{G}^{-1}\right) && (\det M = \det M^{T}) \\
&\implies \det\left(\mathcal{G}_{\mathcal{C}}\right) = (\det \mathcal{G})^{3} \det\left(\mathcal{G}^{-1}\right) && (\det(cM) = c^{n}\det(M), \text{ where } M \text{ is } n \times n) \\
&\implies \det\left(\mathcal{G}_{\mathcal{C}}\right) = (\det \mathcal{G})^{3} (\det \mathcal{G})^{-1} && (\det M^{-1} = (\det M)^{-1}) \\
&\implies \det\left(\mathcal{G}_{\mathcal{C}}\right) = (\det \mathcal{G})^{2} = \boxed{16}.
\end{aligned}$$

13. $\boxed{\textbf{D}}$: $\vec{u} \times \vec{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -1 & 3 \\ -2 & 5 & 1 \end{vmatrix} = \begin{vmatrix} -1 & 3 \\ 5 & 1 \end{vmatrix} \hat{\imath} - \begin{vmatrix} 2 & 3 \\ -2 & 1 \end{vmatrix} \hat{\jmath} + \begin{vmatrix} 2 & -1 \\ -2 & 5 \end{vmatrix} \hat{k} = -16\hat{\imath} - 8\hat{\jmath} + 8\hat{k} = \boxed{\langle -16, -8, 8 \rangle}.$

14. $\boxed{\textbf{C}}$: We can distribute the cross product, giving us $\|(\vec{u} - \vec{w}) \times \vec{v}\| = \|\vec{u} \times \vec{v} - \vec{w} \times \vec{v}\|$. Recall that

$$\vec{w} = \operatorname{proj}_{\vec{v}}(\vec{u}) = \left(\frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}\right)\vec{v}.$$

Observe that, from this, $\vec{w}$ is parallel to $\vec{v}$, meaning that $\vec{w} \times \vec{v} = \vec{0}$. So $\|(\vec{u} - \vec{w}) \times \vec{v}\| = \|\vec{u} \times \vec{v} - \vec{w} \times \vec{v}\| = \|\vec{u} \times \vec{v}\|$. Therefore, $\|(\vec{u} - \vec{w}) \times \vec{v}\| = \|\vec{u} \times \vec{v}\| = \sqrt{(-16)^2 + (-8)^2 + 8^2} = \boxed{8\sqrt{6}}$.

Note: You can also think of $\vec{w}$ as component of $\vec{u}$ in the direction of $\vec{v}$ and $\vec{u} - \vec{w}$ as the component of $\vec{u}$ normal to $\vec{v}$.

15. $\boxed{\textbf{C}}$: From the given, we have that $\langle -4, -3, 7 \rangle_{\mathcal{G}} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} -4 \\ -3 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 21 \end{bmatrix} = \boxed{\langle 0, -5, 21 \rangle}.$

16. $\boxed{\textbf{A}}$: We have that $\langle \alpha, \beta, \gamma \rangle_{\mathcal{G}} = \langle 5, -5, 9 \rangle$. Thus, $\begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 9 \end{bmatrix}$. The last row tells us that $\gamma = 3$, and solving for the rest of the system,

$$\begin{aligned}
\begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \end{bmatrix} &\implies \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 25 \\ -25 \end{bmatrix} && \text{(Multiply Rows 1 and 2 by 5)} \\
&\implies \begin{bmatrix} 9 & -12 \\ 16 & 12 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 75 \\ -100 \end{bmatrix} && \text{(Multiply Row 1 by 3 and Row 2 by 4)} \\
&\implies \begin{bmatrix} 25 & 0 \\ 16 & 12 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -25 \\ -100 \end{bmatrix} && \text{(Add Row 2 to Row 1)} \\
&\implies \begin{bmatrix} 1 & 0 \\ 16 & 12 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -1 \\ -100 \end{bmatrix} && \text{(Multiply Row 1 by } \tfrac{1}{25}\text{)} \\
&\implies \begin{bmatrix} 1 & 0 \\ 0 & 12 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -1 \\ -84 \end{bmatrix} && \text{(Add } -16 \text{ times Row 1 to Row 2)} \\
&\implies \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -1 \\ -7 \end{bmatrix} && \text{(Multiply Row 2 by } \tfrac{1}{12}\text{)}
\end{aligned}$$

Thus, we have $(\alpha, \beta, \gamma) = (-1, -7, 3)$, so $\alpha + \beta + \gamma = \boxed{-5}$.

17. $\boxed{\textbf{D}}$: Observe that

$$\begin{bmatrix} \vec{g}_1 & \vec{g}_2 & \vec{g}_3 \end{bmatrix} = \mathcal{G} \begin{bmatrix} 17 & -16 & 26 \\ -5 & 8 & -13 \\ 0 & 4 & -8 \end{bmatrix}$$

Thus, $\det \begin{bmatrix} \vec{g}_1 & \vec{g}_2 & \vec{g}_3 \end{bmatrix} = \det \left(\mathcal{G} \begin{bmatrix} 17 & -16 & 26 \\ -5 & 8 & -13 \\ 0 & 4 & -8 \end{bmatrix} \right) = (\det \mathcal{G}) \left(\det \begin{bmatrix} 17 & -16 & 26 \\ -5 & 8 & -13 \\ 0 & 4 & -8 \end{bmatrix} \right) = (3)(7) = \boxed{21}$.

18. $\boxed{\textbf{C}}$: If a linear transformation $\mathcal{G}_\tau$, defined via a matrix $\mathcal{G}$, is applied onto a region $\mathcal{R}$ with volume $V_{\mathcal{R}}$ to create a new region $\mathcal{R}'$, then the volume of $\mathcal{R}'$ is equal to $V_{\mathcal{R}}(\det \mathcal{G})$.

The distance between the given vertices is equal to

$$\sqrt{[-25-(-23)]^2 + (10-11)^2 + [-6-(-8)]^2} = \sqrt{(-2)^2 + (-1)^2 + (2)^2} = 3.$$

Thus, the volume of $\mathcal{R} = 3^3 = 27$, so the volume of $\mathcal{R}_{\mathcal{G}}$ is equal to

$$27(\det \mathcal{G}) = 27(3) = \boxed{81}.$$

19. $\boxed{\textbf{A}}$: Let $Q = \mathcal{G}$. Note that Q is orthogonal, that is, $QQ^T = I$, and that $\det Q = 1$. While we could do the basic, albeit time consuming, tactic of computing $\langle -4, 3, 2 \rangle_{\mathcal{G}}$ and $\langle 3, -6, 1 \rangle_{\mathcal{G}}$ then taking the cross product, there is a cleaner solution.

For any two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$, then $\forall \vec{w} \in \mathbb{R}^n$, $\vec{u} \cdot \vec{w} = \vec{v} \cdot \vec{w} \implies \vec{u} = \vec{v}$. We'll use this fact to prove that for orthogonal matrices Q, if $\det Q = 1$, then for $\vec{u}, \vec{v} \in \mathbb{R}^3$,

$$Q(\vec{u} \times \vec{v}) = Q\vec{u} \times Q\vec{v}$$

Let $\vec{w} \in \mathbb{R}^3$. By properties of orthogonal vectors, we have

$$\begin{aligned} Q(\vec{u}\times\vec{v})\cdot Q\vec{w} &= (\vec{u}\times\vec{v})\cdot\vec{w} \\ &= \det\begin{bmatrix}\vec{u} & \vec{v} & \vec{w}\end{bmatrix}^T && \text{(Definition of Triple Scalar Product)} \\ &= \begin{vmatrix}\vec{u} & \vec{v} & \vec{w}\end{vmatrix} && \text{(Determinant of the Transpose)} \\ &= \det Q\begin{vmatrix}\vec{u} & \vec{v} & \vec{w}\end{vmatrix} && (\det Q = 1) \\ &= \begin{vmatrix}Q\vec{u} & Q\vec{v} & Q\vec{w}\end{vmatrix} && \text{(Idea Used in Question 18)} \\ &= (Q\vec{u}\times Q\vec{v})\cdot Q\vec{w} && \text{(Definition of Triple Scalar Product)} \end{aligned}$$

Since we have that $Q(\vec{u}\times\vec{v})\cdot Q\vec{w} = (Q\vec{u}\times Q\vec{v})\cdot Q\vec{w}$, we can conclude that

$$Q(\vec{u}\times\vec{v}) = Q\vec{u}\times Q\vec{v}.$$

Thus, $\langle -4,3,2\rangle_{\mathcal{G}} \times \langle 3,-6,1\rangle_{\mathcal{G}} = (\langle -4,3,2\rangle \times \langle 3,-6,1\rangle)_{\mathcal{G}} = \langle 15,10,15\rangle_{\mathcal{G}} = \boxed{\langle 1,18,15\rangle}$.

Note: $\langle -4,3,2\rangle_{\mathcal{G}} = \langle -\frac{24}{5}, -\frac{7}{5}, 2\rangle$ and $\langle 3,-6,1\rangle_{\mathcal{G}} = \langle \frac{33}{5}, -\frac{6}{5}, 1\rangle$. Though it would result in the same answer, computing the cross product of these two is much more tedious.

20. $\boxed{\text{B}}$: The distance between a point Q and a line $\ell : \vec{u} + t\vec{v}$ is

$$\frac{\|(\vec{PQ})\times\vec{v}\|}{\|\vec{v}\|},$$

where P is a point on ℓ.

In our scenario, choose $P = (3,0,3)$. Then, we have that the distance d between Jason and $\mathcal{H}$ is

$$\begin{aligned} d &= \frac{\|(\langle -4,2,-6\rangle - \langle 3,0,3\rangle)\times\langle 5,-1,7\rangle\|}{\|\langle 5,-1,7\rangle\|} \\ &= \frac{\|\langle -7,2,-9\rangle\times\langle 5,-1,7\rangle\|}{\sqrt{75}} \\ &= \frac{\|\langle 5,4,-3\rangle\|}{5\sqrt{3}} \\ &= \frac{5\sqrt{2}}{5\sqrt{3}} \\ &= \boxed{\frac{\sqrt{6}}{3}}. \end{aligned}$$

21. $\boxed{\text{E}}$: Let $M = \begin{bmatrix} 6 & 1 \\ 11 & -4 \\ -1 & 7 \end{bmatrix}$ and $N = \begin{bmatrix} 5 & -9 & -2 \\ 10 & -3 & 0 \end{bmatrix}$. Note that $\text{rank}(M) = \text{rank}(N) = 2$. For $A \in \mathbb{R}^{x\times y}$ and $B \in \mathbb{R}^{y\times z}$, $\text{rank}(AB) \le \min(\text{rank}(A), \text{rank}(B))$, so $\text{rank}(MN) \le \min(2,2) = 2$. But MN is 3×3, and since the rank of MN is less than the dimension of MN, $\det MN = 0$, $\boxed{\text{NOTA}}$.

22. $\boxed{\text{A}}$: Let $\vec{u}_\ell$ be the reflection of a point $\vec{u}$ over a line $\ell : t(\vec{v})$. It may be useful to draw a diagram to understand this,

$$\vec{u}_\ell = 2\text{proj}_{\vec{v}}(\vec{u}) - \vec{u}.$$

But note that our line does not pass through the origin, so we're unable to directly apply our formula. To account for this, we'll shift both the point we're reflecting and the line by $-\langle 7,-7,7\rangle$, then shift it back by $\langle 7,-7,7\rangle$ once we apply our formula.

Thus, we have that $\vec{u} = \langle 9,3,1\rangle - \langle 7,-7,7\rangle = \langle 2,10,-6\rangle$ and $\vec{v} = \langle 2,2,1\rangle$. Therefore,

$$\begin{aligned}
\vec{u}_\ell &= 2\left(\frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v}\right) - \vec{u} && \text{(Projection Formula)}\\
&= 2\left(\frac{\langle 2,10,-6\rangle\cdot\langle 2,10,-6\rangle}{\langle 2,2,1\rangle\cdot\langle 2,2,1\rangle}\langle 2,2,1\rangle\right) - \langle 2,10,-6\rangle && \text{(Substitution)}\\
&= 2\left(\frac{18}{9}\langle 2,2,1\rangle\right) - \langle 2,10,-6\rangle && \text{(Dot Product)}\\
&= \langle 8,8,4\rangle - \langle 2,10,-6\rangle && \text{(Scalar Multiplication)}\\
&= \langle 6,-2,10\rangle && \text{(Subtraction)}
\end{aligned}$$

Now, adding $\langle 7,-7,7\rangle$ to compensate for our previous shift, we have that the reflection of the point $(9,3,1)$ over the line $\ell : \langle 7,-7,7\rangle + t\langle 2,2,1\rangle$ is $\langle 6,-2,10\rangle + \langle 7,-7,7\rangle \implies \boxed{(13,-9,17)}$.

23. $\boxed{\textbf{B}}$: For the series of matrix powers to converge, the eigenvalues of the matrix must be between -1 and 1. Solving for the eigenvalues of $\begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix}$,

$$\begin{aligned}
\begin{vmatrix} \frac{5}{6}-\lambda & -\frac{1}{6} \\ 1 & 0-\lambda \end{vmatrix} = 0 &\implies \left(\frac{5}{6}-\lambda\right)(-\lambda) + \frac{1}{6} = 0\\
&\implies \lambda^2 - \frac{5}{6}\lambda + \frac{1}{6} = 0\\
&\implies \left(\lambda - \frac{1}{2}\right)\left(\lambda - \frac{1}{3}\right) = 0\\
&\implies (\lambda_1,\lambda_2) = \left(\frac{1}{2},\frac{1}{3}\right)
\end{aligned}$$

Thus, the series will converge. Knowing this, let $R = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix}$ and $S = \sum_{g=0}^{\infty} R^g$. Then,

$$\begin{aligned}
S &= I + R + R^2 + R^3 + \cdots\\
RS &= R + R^2 + R^3 + \cdots\\
&\implies S - RS = I\\
&\implies S(I-R) = I\\
&\implies S = (I-R)^{-1}
\end{aligned}$$

Calculating $S = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix}\right)^{-1} = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} \\ -1 & 1 \end{bmatrix}^{-1}$,

$$\begin{aligned}
S &= \begin{bmatrix} \frac{1}{6} & \frac{1}{6} \\ -1 & 1 \end{bmatrix}^{-1}\\
&= \frac{1}{\left(\frac{1}{6}\right)(1) - \left(\frac{1}{6}\right)(-1)} \begin{bmatrix} 1 & -\frac{1}{6} \\ 1 & \frac{1}{6} \end{bmatrix}\\
&= 3\begin{bmatrix} 1 & -\frac{1}{6} \\ 1 & \frac{1}{6} \end{bmatrix}\\
&= \boxed{\begin{bmatrix} 3 & -\frac{1}{2} \\ 3 & \frac{1}{2} \end{bmatrix}}.
\end{aligned}$$

24. $\boxed{\text{B}}$: Using the Maclaurin series of e^x, we have $e^{\begin{bmatrix}4 & 1\\0 & 2\end{bmatrix}} = \sum_{n=0}^{\infty} \frac{1}{n!}\begin{bmatrix}4 & 1\\0 & 2\end{bmatrix}^n$.

In order to evaluate this sum, we diagonalize $\begin{bmatrix}4 & 1\\0 & 2\end{bmatrix}$.

The eigenvalues of $\begin{bmatrix}4 & 1\\0 & 2\end{bmatrix}$ and $\lambda_1, \lambda_2 = 4, 2$, so we get that

$$\begin{bmatrix}4 & 1\\0 & 2\end{bmatrix} = \begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}\begin{bmatrix}4 & 0\\0 & 2\end{bmatrix}\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}^{-1}$$

Thus,

$$\begin{bmatrix}4 & 1\\0 & 2\end{bmatrix}^n = \begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}\begin{bmatrix}4 & 0\\0 & 2\end{bmatrix}^n\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}^{-1}$$

Substituting this into our sum,

$$\begin{aligned}
\sum_{n=0}^{\infty} \frac{1}{n}\begin{bmatrix}4 & 1\\0 & 2\end{bmatrix}^n &= \sum_{n=0}^{\infty} \frac{1}{n!}\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}\begin{bmatrix}4 & 0\\0 & 2\end{bmatrix}^n\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}^{-1} && \text{(Substitution)}\\
&= \begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}\left(\sum_{n=0}^{\infty} \frac{1}{n!}\begin{bmatrix}4 & 0\\0 & 2\end{bmatrix}^n\right)\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}^{-1} && \text{(Left and Right Factorize)}\\
&= \begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}\left(\sum_{n=0}^{\infty} \frac{1}{n!}\begin{bmatrix}4^n & 0\\0 & 2^n\end{bmatrix}\right)\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}^{-1} && \text{(n-th Power of a Diagonal Matrix)}\\
&= \begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}\begin{bmatrix}\sum_{n=0}^{\infty} 4^n/n! & 0\\0 & \sum_{n=0}^{\infty} 2^n/n!\end{bmatrix}\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}^{-1} && \text{(Move Sum Inside Matrix)}\\
&= \begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}\begin{bmatrix}e^4 & 0\\0 & e^2\end{bmatrix}\begin{bmatrix}1 & 1\\0 & -2\end{bmatrix}^{-1} && \text{(Maclaurin Series of e^x)}
\end{aligned}$$

Multiplying this out, we get $\boxed{\begin{bmatrix}e^4 & \frac{e^4-e^2}{2}\\0 & e^2\end{bmatrix}}$.

Note: WolframAlpha has a different, incorrect definition of e^M for square matrices M. Instead of using the Maclaurin Series of e^x to evaluate e^M, WolframAlpha returns a matrix M' where $M'_{ij} = e^{M_{ij}}$.

25. $\boxed{\text{C}}$: From the given, we have the Maclaurin Series of $\ln(1+x)$. $\ln\left(\begin{bmatrix}4 & 9\\-1 & -2\end{bmatrix}\right) = \ln\left(\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} + \begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}\right)$.

This is in the form of our Maclaurin series, plugging it in we have

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}^n = \begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix} - \frac{1}{2}\begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}^2 + \frac{1}{3}\begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}^3 - \cdots$$

Observe that $\begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}^2 = \begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}$. Then, we have $\begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}^n = \begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}$ for $n \geq 2$. Thus,

$$\ln\left(\begin{bmatrix}4 & 9\\-1 & -2\end{bmatrix}\right) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}^n = \boxed{\begin{bmatrix}3 & 9\\-1 & -3\end{bmatrix}}.$$

Note: Plugging this result into the Mclaurin Series of e^x, we get $\begin{bmatrix}4 & 9\\-1 & -2\end{bmatrix}$, confirming our result.

26. $\boxed{\textbf{B}}$: Denote $p_n(G)$ as the probability that the n-th day is "Gooey", $p_n(S)$ as the probability that the n-th day is "Slimy", and $p_n(T)$ as the probability that the n-th day is "Sticky". Further, denote $\vec{v}_n = \begin{bmatrix} p_n(G) \\ p_n(S) \\ p_n(T) \end{bmatrix}$.

Let $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{3} \\ 0 & \frac{1}{6} & \frac{2}{3} \end{bmatrix}$ define the left-stochastic matrix such that $A\vec{v}_n = \vec{v}_{n+1}$. Since 777 days is such a long time, we can assume that $\lim_{n\to\infty} p_n(G) \approx p_{777}(G)$, which works out in our favor since we want to find the answer choice closest to $p_{777}(G)$.

We have that $\vec{s} = \lim_{n\to\infty} \vec{v}_n$ for $s_1 + s_2 + s_3 = 1$ is the steady-state vector of A, which also satisfies $A\vec{s} = \vec{s}$. So $p_{777}(G) \approx \vec{s}_1$.

Also note that $A\vec{s} = \vec{s}$ implies that $\vec{s}$ is the eigenvector of A that corresponds to an eigenvalue of 1.

Thus, we have that

$$\begin{bmatrix} \frac{1}{2}-1 & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{3}-1 & \frac{1}{3} \\ 0 & \frac{1}{6} & \frac{2}{3}-1 \end{bmatrix} \vec{s} = \vec{0} \implies \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{2}{3} & \frac{1}{3} \\ 0 & \frac{1}{6} & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\implies s_1 = s_2,\ s_2 = 2s_3 \qquad \text{(From Row 1 and Row 3)}$$

Thus, $s_1 + s_2 + s_3 = 1 \implies s_1 + (s_1) + \left(\frac{s_1}{2}\right) = 1 \implies \frac{5}{2}s_1 = 1$. So $s_1 = \frac{2}{5} = 0.4$, meaning that $p_{777}(G) \approx \boxed{0.4}$.

27. $\boxed{\textbf{C}}$: We're given that $\mathcal{G}$ is an upper-triangular matrix with 6 non-zero elements, so

$$\mathcal{G} = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

with $a, b, c, e, d, f \neq 0$.

Observe that since $\mathcal{G}$ is triangular. the elements on the diagonal of $\mathcal{G}$ are the eigenvalues of $\mathcal{G}$, a, d, and f.

Additionally, the non-zero elements of $\mathcal{G}$ are distinct, so $a \neq d$, $d \neq f$, and $a \neq f$. Consider the diagonalization of $\mathcal{G}$,

$$\mathcal{G} = P \begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix} P^{-1}$$

for some 3×3 matrix P, then,

$$\left(P \begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix} P^{-1} \right)^3 = 2 \left(P \begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix} P^{-1} \right)^2 + 41 \left(P \begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix} P^{-1} \right) - 42I$$

Using the properties of diagonalization, we have

$$P \begin{bmatrix} a^3 & 0 & 0 \\ 0 & d^3 & 0 \\ 0 & 0 & f^3 \end{bmatrix} P^{-1} = 2P \begin{bmatrix} a^2 & 0 & 0 \\ 0 & d^2 & 0 \\ 0 & 0 & f^2 \end{bmatrix} P^{-1} + 41P \begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix} P^{-1} - 42I$$

Left multiplying by P^{-1} and right multiplying by P,

$$\begin{bmatrix} a^3 & 0 & 0 \\ 0 & d^3 & 0 \\ 0 & 0 & f^3 \end{bmatrix} = 2 \begin{bmatrix} a^2 & 0 & 0 \\ 0 & d^2 & 0 \\ 0 & 0 & f^2 \end{bmatrix} + 41 \begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix} - 42I \qquad \text{(Note that } P^{-1}(-42I)P = -42P^{-1}P = -42I\text{)}$$

Combining all of these into one matrix, we have

$$\begin{bmatrix} a^3 & 0 & 0 \\ 0 & d^3 & 0 \\ 0 & 0 & f^3 \end{bmatrix} = 2\begin{bmatrix} a^2 & 0 & 0 \\ 0 & d^2 & 0 \\ 0 & 0 & f^2 \end{bmatrix} + 41\begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix} - 42I$$

$$\implies \begin{bmatrix} a^3 - 2a^2 - 41a + 42 & 0 & 0 \\ 0 & d^3 - 2d^2 - 41d + 42 & 0 \\ 0 & 0 & f^3 - 2f^2 - 41f + 42 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, a, d, and f are the roots of the polynomial $p(x) = x^3 - 2x - 41 + 42$, meaning that, by Vietta's, $adf = -42$, so

$$\det\left(\mathcal{G}^2\right) = (adf)^2 = (-42)^2 = 1764.$$

Therefore, the sum of the digits of $\det\left(\mathcal{G}^2\right)$ is $1 + 7 + 6 + 4 = \boxed{18}$.

28. $\boxed{\textbf{A}}$: The adjacency matrix of this directed graph is $A = \begin{bmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$, where A_{ij} corresponds to the number of paths from Node i to Node j. From the given table, we have A^3, as shown below, and A^4 can be calculated through simple matrix multiplication

$$A^3 = \begin{bmatrix} 4 & 6 & 6 \\ 8 & 6 & 7 \\ 4 & 10 & 9 \end{bmatrix} \implies A^4 = \begin{bmatrix} 4 & 6 & 6 \\ 8 & 6 & 7 \\ 4 & 10 & 9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 12 & 16 & 16 \\ 12 & 22 & 21 \\ 20 & 22 & 23 \end{bmatrix}$$

Hence, the number of ways for Jesse to start at Node 1 and end at Node 2 after 3 minutes is $(A^3)_{12} = 6$. The number of ways for Jesse to start at Node 2 and end at Node 3 after 4 minutes is $(A^4)_{23} = 21$.

Therefore, the number of ways for Jesse to start at Node 1, be at Node 2 after 3 minutes, and end at Node 3 after a total of 7 minutes have elapsed is $6(21) = \boxed{126}$

29. $\boxed{\textbf{D}}$: Denote $\phi(\vec{v})$ as the sum of the elements of $\vec{v}$ and $\varphi(M)$ as the sum of the elements of M.

Let $A = \begin{bmatrix} \vec{a_1} & \vec{a_2} & \cdots & \vec{a_n} \end{bmatrix}$ and $B = \begin{bmatrix} \vec{b_1} & \vec{b_2} & \cdots & \vec{b_n} \end{bmatrix}^T$ for n-dimensional column vectors $\vec{a_i}$ and $\vec{b_i}$. Finally, denote $AB = C$. We have that

$$\varphi(C) = \sum_{k=1}^{n} \phi\left(\vec{a_k}\right) \phi\left(\vec{b_k}\right).$$

Note that $\mathcal{G}\mathcal{G}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, so the $\varphi(\mathcal{G}\mathcal{G}^{-1}) = \varphi(I) = 4$. Thus, where $\vec{g_i}$ is the i-th column of $\mathcal{G}$ and $\vec{h_i}$ is the i-th row of $\mathcal{G}^{-1}$, $\varphi(\mathcal{G}\mathcal{G}^{-1}) = \sum_{i=1}^{4} \phi\left(\vec{g_i}\right) \phi\left(\vec{h_i}\right) = 4$. Therefore, we have

$$\begin{aligned} \phi(\vec{g_1}) &= 2 + 2 - 5 = -1 \\ \phi(\vec{g_2}) &= 3 - 3 - 3 = -3 \\ \phi(\vec{g_3}) &= 1 - 2 + 2 = 1 \\ \phi(\vec{g_4}) &= 1 - 1 + 6 = 6 \end{aligned}$$

and

$$\begin{aligned}\phi\left(\vec{h_1}\right) &= -4 - \frac{19}{3} - \frac{13}{3} + \frac{1}{3} = -\frac{43}{3}\\ \phi\left(\vec{h_2}\right) &= 4 + 6 + 4 - \frac{1}{3} = \frac{41}{3}\\ \phi\left(\vec{h_3}\right) &= -3 - \frac{16}{3} - \frac{10}{3} + \frac{1}{3} = -\frac{34}{3}\\ \phi\left(\vec{h_4}\right) &= \spadesuit + \clubsuit + \heartsuit + \diamondsuit\end{aligned}$$

Thus,

$$\begin{aligned}\varphi\left(\mathcal{G}\mathcal{G}^{-1}\right) &= (-1)\left(\frac{-43}{3}\right) + (-3)\left(\frac{41}{3}\right) + (1)\left(-\frac{34}{3}\right) + (6)\left(\spadesuit + \clubsuit + \heartsuit + \diamondsuit\right) = 4\\ &\implies \frac{43}{3} - 41 - \frac{34}{3} + 6(\spadesuit + \clubsuit + \heartsuit + \diamondsuit) = 4\\ &\implies 6(\spadesuit + \clubsuit + \heartsuit + \diamondsuit) = 42\end{aligned}$$

Therefore, $\spadesuit + \clubsuit + \heartsuit + \diamondsuit = \boxed{7}$.

30. $\boxed{\textbf{D}}$: The 10×10 matrix is lower triangular, so the determinant is just the product of the elements on the diagonal.

$$(2)(6)(-1)(\ln \pi)\left(\frac{7}{9}\right)\left(\log_\pi \frac{1}{2}\right)\left(\frac{1}{\ln 2}\right)\left(-\frac{1}{14}\right)(1013)(-6) \implies \boxed{4052}$$